# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed and maintained **RF laboratory benches** for application development and performance validation, integrating custom hardware solutions to accelerate IP functional testing.
- Created and improved **test libraries** to support IP-level testing, enhancing the efficiency of validation processes for next-generation technologies.
- Provided technical support to other engineering teams, assisting with debug efforts and troubleshooting complex issues related to RF application performance.
- Built **testbench components** to support the next generation of IP development, ensuring robust and scalable testing infrastructure.
- Maintained and improved current **hardware laboratory environments**, contributing to faster Go-To-Market plans and enhanced debug capabilities.
- Collaborated with cross-functional teams to integrate and validate new RF application designs, ensuring alignment with project timelines and quality standards.
- Containerized core application services using **Docker** and deployed them to **AWS EC2** instances, configuring basic resource limits for predictable performance during testing phases.

## SKILLS

**Languages** – Python, C/C++, SQL

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

**Tools & Libraries** – Mathworks, NI SW Products, R / RStudio, pandas, NumPy, scikit-learn

**Hardware & RF** – RF Application Design, Mixed Signal Architecture, Design Validation, Testbench Development, Laboratory Bench Maintenance

**Methodologies** – Agile/Scrum, Change Management, Incident Response

## PROJECTS

**RF IP Validation Framework Development (HCL Technologies)**

- Engineered a Python-based framework to automate the validation of RF IP functional performance, reducing manual testing effort by approximately 40%.
- Developed custom hardware interfaces using NI SW Products to connect lab equipment with the automated test framework, enabling real-time data acquisition.
- Integrated the framework with Jenkins for continuous integration, ensuring consistent testing and rapid feedback cycles during development.

**Application Performance Monitoring & Debug Environment (HCL Technologies)**

- Built and maintained a Python scripting suite for automating log analysis and data collection from enterprise applications using Splunk and CloudWatch, identifying root causes for ~15% of P1 incidents.
- Developed REST APIs using Java Spring Boot to expose application health metrics, facilitating integration with external monitoring dashboards and improving overall system visibility.
- Managed CI/CD pipelines via Jenkins for deploying monitoring agents and configuration updates to AWS EC2 environments, ensuring high availability of critical services.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)